

DBMS — Lecture 6

Data Manipulation Operations

Learning Objectives

Understand basic data manipulation operations.

Write INSERT, UPDATE, DELETE, and SELECT statements.

Understand how operations change or retrieve table data.

Data Manipulation

Data manipulation operations are used to add, modify, remove, and retrieve records in a database.

INSERT

Adds new tuples/rows to a table.

```
INSERT INTO Student VALUES (101, 'Raju', 'CSE');
```

UPDATE

Modifies existing records that satisfy a condition.

```
UPDATE Student SET Branch='IT' WHERE Student_ID=101;
```

DELETE

Removes records that satisfy a condition.

```
DELETE FROM Student WHERE Student_ID=101;
```

SELECT

Retrieves required data from one or more tables.

```
SELECT Name, Branch FROM Student WHERE Branch='CSE';
```

Comparison

Operation	Purpose
INSERT	Add new records
UPDATE	Modify existing records
DELETE	Remove records
SELECT	Retrieve records

Important Point

Always use an appropriate **WHERE** condition with UPDATE and DELETE when only selected rows should be affected.

SELECT can use filtering, sorting, grouping, joins, and other query features.

Key Takeaway

Data manipulation operations allow users to create, change, remove, and retrieve the data stored in database tables.